AN EXPLICIT NEGATIVELY ASSOCIATED DETERMINANTAL POINT PROCESS WITH NO SYMMETRIC KERNEL, MINIMAL AT $n = 3$

ABSTRACT. In the discussion section of [9] Poinas asks: *does a negatively associated determinantal point process that does not have the same distribution as a* DPP *with a symmetric kernel even exist?* Read as a question about negative association in the standard sense recalled in Section 1, it does, already on three points. The rational kernel

$$W = \begin{pmatrix} 1/2 & 1/10 & 1/25 \\ 1/10 & 1/2 & 1/10 \\ 1/4 & 1/10 & 1/2 \end{pmatrix}$$

is a DPP kernel, $\mathrm{DPP}(W)$ is negatively associated – strictly so for nonconstant increasing functions on nonempty disjoint supports – and no real symmetric, indeed no Hermitian, kernel has its law, because the law invariant $\sigma^2 - 4uvw$ equals $441/10^8 > 0$ while every real symmetric kernel forces $\sigma^2 - 4uvw = 0$ and every Hermitian kernel forces $\sigma^2 - 4uvw \leq 0$. No smaller ground set carries such a process: at $n \leq 2$ a short case check shows every negatively associated DPP to be symmetrisable. Two qualifications belong here and not further down. First, $\mathrm{DPP}(W)$ is *not* strongly Rayleigh, while the sentence motivating the question in [9] attributes negative association to symmetric-kernel DPPs by citing [2, 7], and the theorem of [2] gives those DPPs the stronger strongly Rayleigh property; nothing here settles the question with negative association replaced by that property. Second, non-symmetrisable determinantal laws are not new: Section 6 records two published constructions, one of them in [9] itself, whose three-point laws already have positive invariant, and for neither is negative association – the other half of the question – established.

1. The question

Fix $n \geq 1$ and $K \in M_n(\mathbb{R})$. Following [9, §2], the *determinantal measure* with kernel K is the set function

$$\mu(\{X : A \subseteq X\}) = \det K_A \qquad (A \subseteq [n]), \tag{1}$$

where K_A is the principal submatrix on A and $\det K_\emptyset = 1$. Its atoms are recovered by Möbius inversion,

$$\mathbb{P}(X = A) = \sum_{A \subseteq T \subseteq [n]} (-1)^{|T|-|A|} \det K_T, \tag{2}$$

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and K is called a DPP *kernel* exactly when all 2^n of these numbers are nonnegative, i.e. when μ is a probability measure; DPP(K) then denotes the resulting random subset of $[n]$. For *symmetric* K, [9, §2] records that this happens iff every eigenvalue of K lies in $[0, 1]$.

A random subset $X \subseteq [n]$ is *negatively associated* (NA) if for all *disjoint* $A, B \subseteq [n]$ and all increasing $f\colon 2^A \to \mathbb{R}$, $g\colon 2^B \to \mathbb{R}$,

$$\mathbb{E}\big[f(X \cap A)\, g(X \cap B)\big] \le \mathbb{E}\big[f(X \cap A)\big]\, \mathbb{E}\big[g(X \cap B)\big]. \tag{3}$$

Negative association is not defined in [9]; it is imported there by the citations [2, 7], and (3) is the standard finite-ground-set form. Every DPP with a symmetric kernel is strongly Rayleigh [2] and hence NA. Two laws DPP(K), DPP(S) coincide iff $\det K_A = \det S_A$ for *every* $A \subseteq [n]$, because by (1) the principal minors are the inclusion probabilities.

The question, and where it is. The paragraph below is quoted from the subsubsection *Negative association* of the section *Discussion and open problems* of [9]. The question is unnumbered and in prose.

> "All DPPs with symmetric kernels satisfy a strong dependency property called negative association [Borcea, Lyons] that extends to continuous DPPs with numerous applications [Ghosh, Poinas17]. Since negative association implies pairwise repulsion then a necessary condition for a DPP with kernel K to be negatively associated is that $K_{i,j}K_{j,i} \ge 0$ for any $i, j \in [n]$ which is obviously not satisfied by all generic DPP kernels. The question naturally arising from this conclusion is what condition on K is needed for the associated DPP to be negatively associated? In fact, does a negatively associated DPP that does not have the same distribution as a DPP with symmetric kernel even exist?"

In the source the four bracketed items are `\cite` commands; apart from those the paragraph is quoted as it stands. The clause settled below is its final sentence.

It is an existence question about *negative association* as defined in (3), and that is the reading taken here. A caution is owed: the paragraph opens by attributing negative association to symmetric-kernel DPPs, and those satisfy the stronger strongly Rayleigh property as recorded above, so a reader may take the question to be about strongly Rayleigh measures instead. The witness below is *not* strongly Rayleigh (Section 7), so under that stronger reading the question is not settled here. The *first* question of the quoted paragraph – "what condition on K is needed?" – is a different target and is not addressed below.

2. The witness

Theorem 1. *Let*

$$W = \begin{pmatrix} 1/2 & 1/10 & 1/25 \\ 1/10 & 1/2 & 1/10 \\ 1/4 & 1/10 & 1/2 \end{pmatrix}.$$

Then (i) W *is a* DPP *kernel;* (ii) DPP(W) *is negatively associated, and* (3) *is* strict *whenever* A, B *are nonempty and* f, g *are both nonconstant; and* (iii) *no real symmetric matrix, and no Hermitian matrix, has the same principal minors as* W*, so* DPP(W) *is the law of no symmetric-kernel and of no Hermitian-kernel* DPP*. In particular the final question of* [9, §Discussion and open problems, "Negative association"]*, read as a question about* (3)*, has the answer* ***yes****.*

Proof of (i). Every 1×1 minor is $1/2$. Every 2×2 principal minor is $1/4 - 1/100 = 6/25$, because $W_{12}W_{21} = W_{13}W_{31} = W_{23}W_{32} = 1/100$ (namely $\frac{1}{10}\frac{1}{10}$, $\frac{1}{25}\frac{1}{4}$, $\frac{1}{10}\frac{1}{10}$). By the six-term Laplace expansion,

$$\det W = \tfrac{1}{8} - 3 \cdot \tfrac{1}{2} \cdot \tfrac{1}{100} + W_{12}W_{23}W_{31} + W_{13}W_{32}W_{21} = \tfrac{1}{8} - \tfrac{3}{200} + \tfrac{1}{400} + \tfrac{1}{2500} = \tfrac{1129}{10000}.$$

The minors depend only on $|A|$, so the law is exchangeable. By (2) the eight atoms are

$$\begin{aligned} \mathbb{P}(X = \emptyset) &= 1 - 3 \cdot \tfrac{1}{2} + 3 \cdot \tfrac{6}{25} - \tfrac{1129}{10000} = \tfrac{1071}{10000}, \\ \mathbb{P}(X = \{i\}) &= \tfrac{1}{2} - 2 \cdot \tfrac{6}{25} + \tfrac{1129}{10000} = \tfrac{1329}{10000}, \\ \mathbb{P}(X = \{i, j\}) &= \tfrac{6}{25} - \tfrac{1129}{10000} = \tfrac{1271}{10000}, \\ \mathbb{P}(X = [3]) &= \tfrac{1129}{10000}. \end{aligned}$$

Their sum is $(1071 + 3 \cdot 1329 + 3 \cdot 1271 + 1129)/10000 = 1$ exactly, and all eight are *strictly* positive, the smallest being $1071/10000$. Hence μ is a probability measure and W is a DPP kernel. (Also $\mathbb{P}(X = \emptyset) = \det(I - W) \neq 0$, so $L = W(I - W)^{-1}$ exists and DPP(W) is an L-ensemble.) □

Proof of (ii). Write $D(f, g) = \mathbb{E}[fg] - \mathbb{E}[f]\mathbb{E}[g]$, so that (3) reads $D(f, g) \leq 0$. The defect D is bilinear and vanishes as soon as f or g is constant; in that case (3) holds with equality, which is why strictness is asserted only for nonconstant f, g. Every increasing f on a finite Boolean lattice is a *nonnegative* combination of up-set indicators plus a constant, namely $f = v_1 + \sum_{i \geq 2}(v_i - v_{i-1})\mathbf{1}_{\{f \geq v_i\}}$ over its distinct values $v_1 < \cdots < v_m$; if f is nonconstant then $m \geq 2$, each coefficient $v_i - v_{i-1}$ is strictly positive, and each $\{f \geq v_i\}$, $i \geq 2$, is a proper nonempty up-closed family. So (3) for all increasing f, g is *equivalent* to the finite list of inequalities $D(\mathbf{1}_U, \mathbf{1}_V) \leq 0$ over proper nonempty up-closed $U \subseteq 2^A$, $V \subseteq 2^B$, not merely implied by it, and if every one of those is strict then $D(f, g) < 0$ for every pair of nonconstant increasing f, g.

At $n = 3$ the disjoint pairs are $(\{i\}, \{j\})$ and $(\{i\}, \{j, k\})$, three of each; the proper nonempty up-sets number 1 on a one-element lattice and 4 on a two-element lattice, so there are $3 \cdot 1 \cdot 1 + 3 \cdot 1 \cdot 4 = 15$ inequivalent inequalities, 30 when both orders of each pair are counted. Exchangeability collapses them to three classes, each *strict*:

class	LHS	RHS	slack
$\{i \in X\}$ vs $\{j \in X\}$	$6/25$	$1/4$	$1/100$
$\{i \in X\}$ vs $\{j, k \in X\}$	$1129/10000$	$3/25$	$71/10000$
$\{i \in X\}$ vs $\{X \cap \{j, k\} \neq \emptyset\}$	$3671/10000$	$19/50$	$129/10000$

All 30 ordered inequalities hold, with minimum slack $71/10000 > 0$. (The first class is exactly the necessary condition $K_{ij}K_{ji} \geq 0$ of [9] quoted in Section 1; here all three products equal $1/100$.) ☐

For (iii) we isolate the invariant.

3. The obstruction is a single law invariant

For a triple $\{i, j, k\} \subseteq [n]$ and $K \in M_n(\mathbb{C})$ put

$$\begin{aligned} u &= K_{ij}K_{ji}, \qquad v = K_{ik}K_{ki}, \qquad w = K_{jk}K_{kj}, \\ p &= K_{ij}K_{jk}K_{ki}, \qquad q = K_{ik}K_{kj}K_{ji}, \qquad \sigma = p + q. \end{aligned} \tag{4}$$

Lemma 2. *$pq = uvw$ identically, with no hypothesis on K; consequently $\sigma^2 - 4uvw = (p - q)^2$. Moreover u, v, w and σ are functions of the principal minors of K alone:*

$$u = K_{ii}K_{jj} - \det K_{\{i,j\}}, \qquad \sigma = \det K_{\{i,j,k\}} - K_{ii}K_{jj}K_{kk} + K_{ii}w + K_{jj}v + K_{kk}u.$$

Proof. $pq = (K_{ij}K_{ji})(K_{jk}K_{kj})(K_{ki}K_{ik}) = uvw$ after regrouping the six factors, and $(p + q)^2 - 4pq = (p - q)^2$. The first displayed identity is the 2×2 expansion; the second is the six-term expansion of $\det K_{\{i,j,k\}}$ with $p + q$ collected. ☐

Theorem 3 (representability)**.** *Fix $u, v, w > 0$ and a value σ. Among matrices with the invariants* (4) *on a given triple: a real matrix exists iff $\sigma^2 \geq 4uvw$; a real **symmetric** one exists iff $\sigma^2 = 4uvw$; a **Hermitian** one exists iff $\sigma^2 \leq 4uvw$.*

Proof. By Lemma 2, p and q are the two roots of $z^2 - \sigma z + uvw$. For real matrices they must be real, i.e. $\sigma^2 \geq 4uvw$. For real *symmetric* S the two 3-cycle products coincide, $p = q = S_{ij}S_{jk}S_{ki}$, so $\sigma = 2p$ and $uvw = p^2$, i.e. $\sigma^2 = 4uvw$; conversely $\sigma^2 = 4uvw$ with $u, v, w > 0$ is realised by $S_{ij} = \varepsilon_{ij}\sqrt{u}$ etc. with signs chosen so that the product is $\sigma/2$. For Hermitian S one has $q = \bar{p}$, so $\sigma = 2\operatorname{Re} p$ and $uvw = |p|^2$, whence

$$\sigma^2 - 4uvw = 4(\operatorname{Re} p)^2 - 4|p|^2 = -4(\operatorname{Im} p)^2 \leq 0; \tag{5}$$

conversely any σ with $\sigma^2 \leq 4uvw$ is realised by a p on the circle $|p|^2 = uvw$ with $2\operatorname{Re} p = \sigma$. ☐

Proof of Theorem 1(iii). For W on $\{1,2,3\}$: $u = v = w = 1/100$, $p = W_{12}W_{23}W_{31} = 1/400$, $q = W_{13}W_{32}W_{21} = 1/2500$, so $\sigma = 29/10000$ and, by Lemma 2,

$$\sigma^2 - 4uvw = (p-q)^2 = \left(\tfrac{1}{400} - \tfrac{1}{2500}\right)^2 = \left(\tfrac{21}{10000}\right)^2 = \frac{441}{10^8} > 0.$$

Since u, v, w, σ are functions of the principal minors alone, any matrix with the law of $\mathrm{DPP}(W)$ has the same value $441/10^8$; by Theorem 3 that excludes every real symmetric matrix (which needs 0) and, by (5), every Hermitian matrix (which needs ≤ 0). $\square$

Remark 4 (an algebra-free version of the same exclusion). A real symmetric S with the principal minors of W must have $S_{ii} = 1/2$ and $S_{ij}^2 = W_{ii}W_{jj} - \det W_{\{i,j\}} = 1/100$, hence $S_{ij} = \pm 1/10$: the modulus is *forced*, and $1/100$ is a rational square, so no irrationality escapes. Enumerating the eight sign patterns gives only two values of $\det S$, namely $14/125$ and $27/250$, and neither is $\det W = 1129/10000$. This settles (iii) for the real symmetric case with no algebra at all.

4. Minimality in n

Theorem 5. *A negatively associated* DPP *whose law is the law of no symmetric-kernel* DPP *exists for* ***exactly*** *the* $n \geq 3$. *In particular* $n = 3$ *is the smallest possible ground set.*

Proof. $n \geq 3$. Take $K_n = W \oplus \mathrm{diag}(c_4, \ldots, c_n)$ with rationals $c_i \in (0,1)$. For block-diagonal K the minors factorise over blocks – this is [9, §2]: "if K is a block diagonal matrix with diagonal blocks $K_1, \ldots, K_l$ then K being a DPP kernel is equivalent to each K_i being a DPP kernel" – so the atoms factorise, all stay strictly positive, and $\mathrm{DPP}(K_n)$ is the independent superposition of $\mathrm{DPP}(W)$ with independent Bernoulli(c_i)'s. Negative association is closed under independent superposition [5, property P7], so $\mathrm{DPP}(K_n)$ is NA. The obstruction is inherited by the principal submatrix on $\{1,2,3\}$: any symmetric or Hermitian L with the same law has the same principal minors, hence the same u, v, w, σ on $\{1,2,3\}$, hence $\sigma^2 - 4uvw \leq 0$ there, against $441/10^8 > 0$. No information about L outside $\{1,2,3\}$ is used, so one bad triple obstructs at every larger n.

$n \leq 2$ *is impossible.* At $n = 1$, (3) is vacuous (there is no pair of disjoint nonempty subsets) and $K = (K_{11})$ is symmetric anyway. At $n = 2$ the law is determined by (K_{11}, K_{22}, u) with $u = K_{12}K_{21}$, and the single NA inequality is $\mathbb{P}(1,2 \in X) \leq \mathbb{P}(1 \in X)\mathbb{P}(2 \in X)$, i.e. $K_{11}K_{22} - u \leq K_{11}K_{22}$, i.e. $u \geq 0$. Then the symmetric $S = \left(\begin{smallmatrix} K_{11} & \sqrt{u} \\ \sqrt{u} & K_{22} \end{smallmatrix}\right)$ reproduces both 1×1 minors and the 2×2 minor $K_{11}K_{22} - u$, hence every atom; and S has spectrum in $[0,1]$ because its characteristic polynomial satisfies $\chi(0) = \mathbb{P}(X = \{1,2\}) \geq 0$, $\chi(1) = \mathbb{P}(X = \emptyset) \geq 0$ and $0 \leq \frac{1}{2}\operatorname{tr} S \leq 1$. So every NA DPP on at most two points is symmetrisable. $\square$

5. Real-symmetric-kernel laws lie on a hypersurface

Coordinatise the 3-point DPP laws by the 7 invariants $(a_1, a_2, a_3; u, v, w; \sigma)$ with $a_i = K_{ii}$; by Lemma 2 these are exactly the principal minors, so they are exactly the law.

Theorem 6. *The negatively associated* 3*-point* DPP *laws have nonempty interior in this* 7*-dimensional space, whereas the laws of real symmetric kernels are confined to the hypersurface* $\sigma^2 = 4uvw$*, a closed Lebesgue-null set of codimension one. Consequently* ***Lebesgue-almost every negatively associated*** **3*-point*** **DPP** ***law is the law of no*** **DPP** ***with a real symmetric kernel.***

The restriction to real symmetric kernels is essential: Hermitian kernels are confined only to the region $\sigma^2 \le 4uvw$, which is full-dimensional, so no analogous genericity statement against Hermitian kernels is made here; see Section 7.

Proof. At the law of W all eight atom inequalities, all 30 NA inequalities, $u, v, w > 0$ and $\sigma^2 - 4uvw > 0$ hold *strictly* (Section 2), and each is a polynomial in the seven free coordinates; so an open neighbourhood of that law consists of NA DPP laws with $\sigma^2 > 4uvw$. The symmetric locus is $\{\sigma^2 = 4uvw\}$ by Theorem 3, the zero set of a nonzero polynomial. □

6. What was already in print

Determinantal laws that are the law of no symmetric kernel are already in print, in two published constructions; whether either is negatively associated we do not know. We record this because presenting non-symmetrisability as a discovery would be an overclaim.

(a) The asking paper's own coupling. Poinas' coupling construction [9, Prop. "Construction coupling"] builds from a symmetric DPP kernel K and a parameter vector p the $2n$-site kernel $\mathcal{K} = \left(\begin{smallmatrix} K & K-D(p) \\ K & K \end{smallmatrix}\right)$. On the triple $\{1, 2, 1'\}$ (two sites of the first copy, one of the second) one gets $u = w = b^2$ with $b = K_{12}$, $v = K_{11}(K_{11} - p_1)$ and $\sigma^2 - 4uvw = (b^2 p_1)^2 > 0$ whenever $p_1 > 0$ and $b \neq 0$. So the paper's own coupling law is already non-symmetrisable; should it also be negatively associated – which we have not checked – bare existence would follow from that construction alone. Poinas does not test it for negative association.

(b) The one-dependent class the paper itself cites. Borodin, Diaconis and Fulman [3] – reference `[Borodin09]` of [9] – construct determinantal two-block-factor processes with nonsymmetric kernels; their $X_i = \mathbf{1}\{Y_i \le p,\, Y_{i+1} > p\}$ gives, at $n = 3$, $K = d\left(\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{smallmatrix}\right)$ with $d = p(1-p)$, whose law forces $\det K = 0$ while a symmetric competitor would need $-d^3$. Their Theorem 4.1 and Example 6 give $\mathbb{P}(0,0,0) = 1 - 3d + d^2$ at $a_2 = d$, $a_3 = a_4 = 0$ – the same atom we obtain by Möbius inversion from that kernel, an external published checksum. Not every member works: their descent process of a uniform permutation on three sites has $u = w = 1/12$, $v = 0$, $\sigma = 0$, hence $\sigma^2 = 4uvw$, and *is* symmetrisable.

What this leaves. In the sources we read we did not find negative association stated for any nonsymmetric DPP kernel, nor the question whether an NA DPP law can fail to be a symmetric-kernel law: [3] contains no occurrence of "negative association", "negative dependence" or "repulsi", and its only remark on the point is the warning that the negative-dependence references "mostly deal with symmetric kernels... whereas all of our examples involve nonsymmetric kernels". This is a statement about what those sources *say*, not about what is decidable: for the two-block-factor process displayed in (b) the three-point NA inequalities can be tested directly from the atoms recorded there, and we do not present that test as open. Our search could not reach MathSciNet or Google Scholar (Section 7), so no priority is claimed here. What is offered is the explicit witness with its exact verification, the impossibility at $n \le 2$, and the codimension count of Section 5.

Finally, the direction of Theorem 3 used here is the *elementary* one (symmetric $\Rightarrow p = q \Rightarrow \sigma^2 = 4uvw$). The hard converse – when principal minors determine a matrix up to diagonal similarity and transposition – is the subject of [6, 10, 8, 1], and is not needed.

7. What is not settled

- **Theorem 6 has no Hermitian analogue.** The law $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}; \frac{1}{100}, \frac{1}{100}, \frac{1}{100}; 0)$ has atoms $\frac{11}{100}, \frac{13}{100}(\times 3), \frac{13}{100}(\times 3), \frac{11}{100}$, is NA with slack 1/100, and *is* the law of the Hermitian kernel $\begin{pmatrix} 1/2 & 1/10 & -i/10 \\ 1/10 & 1/2 & 1/10 \\ i/10 & 1/10 & 1/2 \end{pmatrix}$; all inequalities being strict, an open 7-dimensional set of NA laws is Hermitian-representable. The Hermitian exclusion in Theorem 1 is a *pointwise* statement about W, not a genericity statement.
- DPP(W) **is not strongly Rayleigh**, so the analogous existence question for strongly Rayleigh measures is *not* answered here. With $f(x) = \sum_A \mathbb{P}(X = A) \prod_{i \in A} x_i$ and $D_{ij} f = \partial_i f\, \partial_j f - f\, \partial_i \partial_j f$, Brändén's criterion for multiaffine polynomials [4] requires $D_{ij} f \ge 0$ on $\mathbb{R}^3$; at $x = (-4, -4, -2)$ and $(i, j) = (1, 2)$ one computes $\partial_1 f = \partial_2 f = 2735/10^4$, $f = -7675/10^4$, $\partial_1 \partial_2 f = -987/10^4$, so $D_{12} f = (2735^2 - 7675 \cdot 987)/10^8 = -19/20000 < 0$. DPP($W$) *is* negatively associated, but any restatement of this result must say "negatively associated", never "strongly Rayleigh".
- **The continuous-space version of the question is untouched.** What is settled here is the finite-ground-set question of [9]; "Continuous setting" is a separate subsubsection of the same discussion section.

- **Prior-art coverage.** MathSciNet and Google Scholar were not reachable to us, so a scoop in an unpublished thesis, in lecture notes, or in a work indexed only by MR cannot be excluded.

8. Reproduction and verification

Every object above is printed in full, in exact rationals, so the entire verification can be redone with a pen: the eight atoms of W by (2), the three slack classes, the six-term determinant, Lemma 2, and Theorem 1(iii) in four multiplications and one subtraction. Remark 4 replaces even Theorem 3 by an eight-case enumeration.

The accompanying program `verify.py` (Python 3.9+, standard library only, exact `Fraction` arithmetic, no float decision anywhere) reads the matrices *as printed above* and re-derives the numbers in the results of this paper, apart from the two groups listed at the end of this section: the minors and atoms of W by two independent routes, negative association brute-forced *from* (3) over all ordered pairs of disjoint nonempty sets and all pairs of proper nonempty up-closed families with up-closure *tested* rather than assumed, the invariants of (4) with u and σ cross-checked against the principal minors, the eight-sign-pattern exclusion of Remark 4, the identity (5) on 1728 Hermitian candidates with the forced moduli, the padding and the $n \leq 2$ sweep of Theorem 5, and the strong-Rayleigh point above. Its up-set enumerator is pinned against the Dedekind numbers $2, 3, 6, 20, 168, 7581$ (OEIS A000372) and its test counts against the prediction $30, 348, 4560, 140058$ for $n = 3, 4, 5, 6$. The recorded run reports `VERDICT: ALL 71 CHECKS PASS` and its transcript, together with the scope statement the program prints for itself, accompanies this paper.

Reading the transcript against this paper. Its sections 1 and 2 verify Theorem 1(i) and (ii); its section 3 verifies Lemma 2, Theorem 1(iii) and Remark 4; its section 8, headed "Theorem D", verifies the padding and $n \leq 2$ parts of Theorem 5; its section 10 verifies the strong-Rayleigh point of Section 7. Its remaining headings name objects not discussed above and on which nothing above depends: the controls of its section 4, the closed-form slack formula it calls "Theorem B", a second $n = 3$ witness W', the σ-interval it calls "Theorem C", the $n = 4$ witness, the box certificate of its section 9, and the conditional-association checks. Those labels are the program's own and do not correspond to statements of this paper.

What the program does *not* cover. Two groups of numbers appearing above are outside its 71 checks, and are stated here so that a reader does not look for them in the transcript. They are elementary and were checked by hand only.

- The arithmetic of Section 6: the triple invariants of Poinas' coupling ($u = w = b^2$, $v = K_{11}(K_{11} - p_1)$, gap $(b^2 p_1)^2$), and the Borodin–Diaconis–Fulman numbers ($\det K = 0$ against a symmetric competitor's $-d^3$, the external checksum $\mathbb{P}(0,0,0) = 1 - 3d + d^2$, and $u = w = 1/12$, $v = 0$ for the descent process). The program's "source's own kernel" control is the uniform-on-even-subsets kernel, a *different* object from the two discussed there.

- The Hermitian counterexample in the first item of Section 7 (atoms 11/100 and 13/100, NA slack 1/100). The program verifies the Hermitian *inequality* (5) on 1728 candidates, which is what Theorem 1(iii) needs; it does not evaluate that particular kernel.

Neither group is load-bearing for any theorem above: the first is attribution, the second is a disclaimer that only weakens Theorem 6.

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